Lecture 27, DAA April 24

$x_i \in \{True, False\}$ $1 \leq i \leq n$ boolean variables

$x_i, \overline{x_i}$ are literals

C_j : clause $(x_{j1} \vee x_{j,2} \vee x_{j3} \cdots \vee x_{jK})$

disjunction of K literals

K-CNF Conjunctive Normal Form with K literals

$(x_{11} \vee x_{12} \vee \cdots x_{1K}) \wedge (x_{21} \vee x_{22} \cdots x_{2K}) \wedge \cdots (x_{m1} \vee \cdot x_{mK})$

Can you satisfy $(x_1 \vee \bar{x}_2) \wedge (\bar{x}_1 \vee \bar{x}_2) \wedge (x_1 \vee x_2)$

Cook Levin Theorem: For any problem $\pi \in NP$,

there exists a CNF f with atmost K literals per clause and a polynomial (in the inputsize of π)

number of boolean variables and a polynomial
number of clauses such that
F is satisfiable iff π has an answer Yes
F can be compiled in polynomial time.

– Show that 3CNF is NPC –

Given a CNF F with at most k literals per clause, construct a 3CNF F' such that
F is satisfiable $\Longleftrightarrow$ F' is satisfiable

Simple observation – If all clauses have 3 literals then there is nothing to do

Let's consider ...

- Let's consider clauses of size

(i) Two $(x_1 \vee x_2)$

$\Longleftrightarrow (x_1 \vee x_2 \vee y) \wedge (x_1 \vee x_2 \vee \bar{y})$ y: new variable

(ii) One $(x_1) \Longleftrightarrow$

$(x_1 \vee y_1 \vee y_2) \wedge (x_1 \vee \bar{y}_1 \vee y_2) \wedge (x_1 \vee y_1 \vee \bar{y}_2) \wedge (x_1 \vee \bar{y}_1 \vee \bar{y}_2)$

y_1, y_2 new variables

(iii) Three - no action needed

(iv) $\geqslant 4$, say $(x_1 \vee x_2 \vee x_3 \vee x_4 \vee x_5)$

$(x_1 \vee x_2 \vee y_1) \wedge (\bar{y}_1 \vee x_3 \vee y_2) \wedge (\bar{y}_2 \vee x_4 \vee x_5)$

y_1, y_2 : new additional variables

- Overall how many additional variables are needed

- Count how many additional variables are needed

- How many clauses are there in the 3CNF

The entire transformation is doable by an algorithm in poly-time

Verify that f' is satisfiable $\iff$ f is satisfiable

So $L_{SAT} \leq^{poly} L_{3SAT}$

$\Rightarrow$ For any $\pi \in NP$

$\pi \leq^{poly} L_{SAT} \leq^{poly} L_{3SAT}$

$\Rightarrow$ $\pi \leq^{poly} L_{3SAT}$ $\Rightarrow$ L_{3SAT} is NP hard

Is $L_{3SAT} \in NP$?

L_{VC} : the set of strings that are YES instances of Vertex Cover

Is L_{vc} NPC?

(i) $L_{vc} \in NP$

(ii) Some NPC $\leq^{poly} L_{vc}$?

Consider 3SAT formula $x_{ij} \in \{x_1, \bar{x}_1, x_2, \bar{x}_2 \cdots x_n, \bar{x}_n\}$

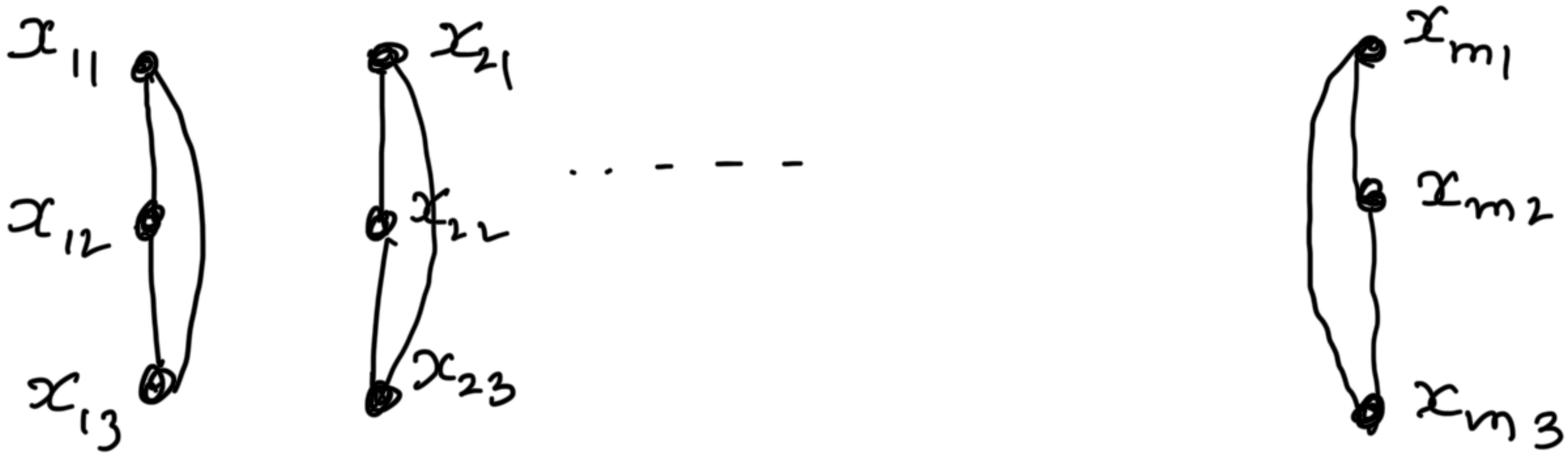
$$F = (x_{11} \vee x_{12} \vee x_{13}) \wedge (x_{21} \vee x_{22} \vee x_{23}) \cdots (x_{m_1} \vee x_{m_2} \vee x_{m_3})$$

We want to construct an instance of V.C.

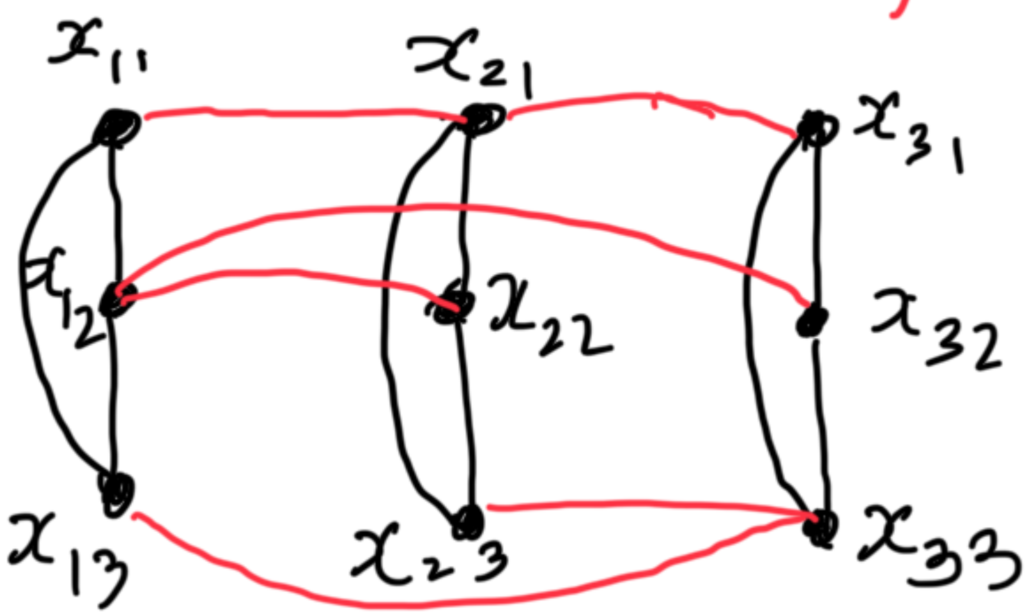
$$\mathcal{I} = (V, E, K) \quad 1 \leq K \leq |V|$$

such that F is satisfiable $\iff$ $\mathcal{I}$ is a Yes instance

(without knowing if F is satisfiable!)



Consider $(x_1 \vee \bar{x}_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee x_3) \wedge (x_1 \vee x_2 \vee \bar{x}_3)$



$K = \ ? \quad 2m$

Claim: F is satisfiable iff $G(F)$ has a VC of size k

(1) F is satisfiable $\Rightarrow$ At least one literal is true in every clause

Choose the remaining two (break ties arbitrarily) in the subset W

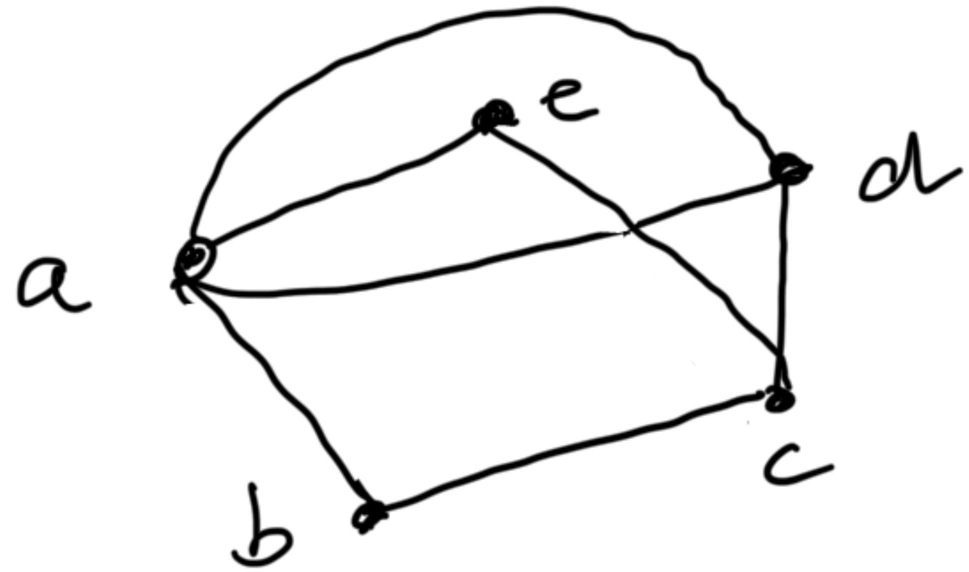
$\boxed{W \text{ is a vertex cover of size } 2m}$

(2) $G(F)$ has a VC of size $2m$ $\Rightarrow$ F is satisfiable

Truth assignment: Choose the variable in a $\triangle$ not in VC to be True. Show this is a consistent assignment

How do we tackle hard problems that cannot be avoided?

Consider the problem of minimum Vertex Cover



① Find a maximal matching

② W = set of endpoints of matching edges

Claim : (i) W is a vertex cover

(ii) $|W| \leq 2 \cdot |O^*|$ O^* : min VC

Other important complexity classes

CO-NP : complement of NP

PSPACE: polynomial space

RP, ZPP, BPP, PP : probabilistic classes
(~~for~~ randomized algorithms)